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ABSTRACT
The selective reduction of low-grade manganese ore followed by magnetic separation was proposed to produce rich-manganese ore. The optimized parameters include a roasting temperature of 1050°C, a roasting time of 6 h, a manganese ore size of 8–13 mm, and an FC/O ratio of 2.5. The reduction roasting products of low-grade manganese ore, mainly comprising of MnO, metallic iron, and residual gangue, could be separated effectively to obtain rich-manganese ore and magnetic product. The RMn, RFe, and TMn can reach the peak values of 71.00%, 93.60%, and 56.20wt%, respectively, at a rational magnetic field strength of 100 mT. In particular, the Mn/Fe ratio is up to 10.85, which meets the requirements of ferromanganese alloy smelting.

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KEYWORDS
Low-grade manganese ore; selective reduction; magnetic separation; thermodynamics

Introduction

Manganese plays a strategically important role in many fields including steel production, chemical, paint, and battery industries. Currently, around 95% of the manganese alloys, mainly produced from manganese ore, have been included as deoxidizer, desulfurizer, and alloy in steelmaking industry.[1–3] Manganese ore with a Mn content of higher than 44wt% is defined as high-grade manganese ore. Generally, high-grade manganese ore is used for direct smelting of ferromanganese alloys. However, high-grade manganese ore is facing a serious scarcity in the global market. The manganese ores with high iron content were unsuitable for direct smelting of ferromanganese alloy. In addition, the iron and manganese in manganese ore cannot be separated by conventional gravity separation and magnetic separation methods mainly due to the narrow density and magnetism of Fe and Mn oxides.[4] The previous works have mainly studied the hydrometallurgical and pyrometallurgical route to upgrade manganese content from low-grade manganese ore. The typical process of hydrometallurgical route applied different reducing agents such as black charcoal,-[5] pyrite,[6] oxalic acid,[7] hydrogen peroxide,[8] and organic biomass.[9–13] However, for the low-grade manganese ore, the complete leaching of the iron in ore generally requires fine grinding and a higher amount of acid which results in an increase in the production cost and a severe environmental pollution.[2]

Many researchers have proposed pyrometallurgical route to separate iron and manganese from the Mn–Fe oxide through magnetic roasting or direct reduction roasting followed by magnetic separation processes.[14–18] Yao et al have concluded that the manganese content in the ore increased from around 36wt% to more than 44wt%, and almost 50 wt% of iron was removed at a Mn loss of around 5wt% after magnetic separation, it difficult to achieve any significant Mn upgrading from low-grade manganese ore.[4] Singh et al. [19] utilized low-grade manganese ore by gravity separation and magnetic separation techniques, and obtained rich-manganese ore containing a TMn of 42wt% and a Mn/Fe of 5 at around a Mn recovery ratio of 48%. Zhang et al. obtained a magnetic concentrate with 86.39 wt% TFe and 96.21 wt% Fe recovery as well as a nonmagnetic product with 84.84 wt% TMn and 85.96 wt% Mn recovery.[3] Among most of previous work on the effective separation and recovery of Mn and Fe from low-grade manganese ore. The final magnetic materials can indirectly meet the requirement of high-quality ferromanganese alloy production.[20] Previous studies on the separation and recovery of iron and manganese from the low-grade manganese ores by reduced gas as reduced agent in the process of magnetic roasting. The research on selective reduction of lump low-grade manganese ore by lump coal as reduced agent is rarely reported. Therefore, it is essential to develop reasonable and suitable techniques for upgrading low-grade manganese ore to efficiently separation and recovering of Mn from the low-grade manganese ore.

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The magnetic roasting and magnetic separation process have been widely used to enhance the effective separation of iron and gangue minerals from the refractory minerals to extract valuable metals.[21–27] This study aimed to investigate the feasibility of Fe and Mn separation from manganese ore via selective reduction followed by magnetic separation process. The impacts of reduction temperature, roasting time duration, the FC/O, ore size on the reduction of manganese ore, and magnetic field strength on the separation efficiency of iron and manganese are investigated. Then, the characterization of reduced product, magnetic and nonmagnetic products was carried out to reveal the function mechanism of selective reduction and magnetic separation.

Experimental

Raw materials

The chemical compositions of manganese ore and industrial compositions of reduction coal used in this study are given in Tables 1 and 2, respectively. The raw manganese ores are of typical low-grade manganese content (29.50wt%), high Fe content (28.28wt%), and low Mn/Fe rate ratio (1.04). Figure 1 shows the X-ray diffraction (XRD) analysis of raw manganese ore. It is evident from the figure that the manganese is mainly comprised of manganite MnO(OH) and bixbyite ($(Fe,Mn)_2O_3$). The iron compounds are mainly found in the form of hematite (Fe_2O_3) and bixbyite ($(Fe,Mn)_2O_3$). The Si, Al, and Ca elements in this ore, which have negative influences on manganese recovery and efficiency of iron-removal, are found mainly in form of quartzite (SiO_2) and chabazite ($Ca_{1.95}Al_{3.9}Si_{8.1}O_{24}$).

Experimental procedure

The raw materials are dried at 105°C for 3 h in a draught drying cabinet. Then the coals used in the study are crushed by a jaw crusher (PE60 × 100 type) to a size of less than 3 mm. In addition, in order to investigate the effect of particle size on the reduction process, the manganese ores with different particle size are prepared. About 200 g manganese ore and reduction coal are adequately mixed in a certain proportion of FC/O. It should be noted that the FC/O is the mole ratio of fixed carbon in coal to reducible oxygen of iron and manganese in manganese ore. Next, the mixtures are charged into a corundum crucible and placed into a heating furnace. The mixed material surface is covered by a coal powder layer with a proportion of 10%.[23] The heating rate of the furnace is 10°C/min in the temperature range below 800°C and then 8°C/min to the reduction temperature with simulated rotary kiln conditions. During the reduction experiment, the effect of reduction temperature and roasting time duration of manganese ore are investigated. When the reduction roasting of manganese ore is completed, the reduced products are taken out and cooled to room temperature in N_2 atmosphere (cooling rate 15°C/min). The reduced manganese ore was grounded by 2-MZ centrifugal grinding machine to 100% passing 74 μm followed by magnetic separation using a Davies Magnetic Tube (DTCXG-ZN50). During the magnetic separation experiment, the influence of magnetic field strength is investigated. After magnetic separation, the Mn and Fe contents of magnetic and nonmagnetic products are measured by chemical titration.

In the experiment process, a preliminary test is first carried out to study the feasibility of magnetic separation for reduced manganese ore. The experiments are mainly conducted to investigate the effects of reduction temperature, roasting time duration, FC/O, manganese ore size, and magnetic field strength to obtain the

Table 2. Industrial analysis of reduced coal for experiment (wt %).

Item	Fixed carbon (FC)	Ash (A_{ad})	Volatile (V_{ad})	Water
Content	52.98	5.30	40.70	1.02

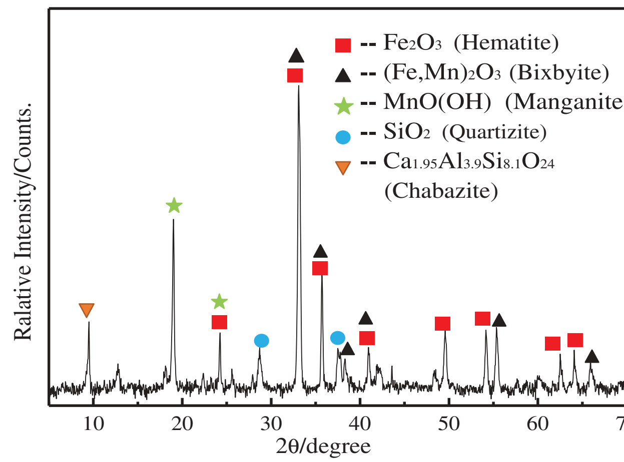


Figure 1. XRD pattern of the original manganese ore.

Table 1. Chemical composition of typical low-grade manganese ore (wt%).

Element	TFe	TMn	FeO	SiO_2	Al_2O_3	CaO	MgO	S	P	Mn/Fe
Content	28.28	29.50	0.09	3.72	7.82	0.28	0.44	0.029	0.049	1.04

complete and rational technological parameters. The base reduction parameters include a roasting time duration of 6 h, a reduction temperature of 1050°C, a manganese ore size of 5–8 mm, an FC/O ratio of 2.5, and a magnetic field strength of 50 mT. In this study, the single-factor control variable methods, where only a single parameter changed and other parameters were maintained in the basic conditions, was used to investigate the effects of various roasting parameters on the recovery of valuable elements in the magnetic separation process.

The recovery ratio of Mn (R_{Mn}), the removed ratio of Fe (R_{Fe}), the mass ratio of Mn to Fe in the nonmagnetic (Mn/Fe), the total content of Mn (T_{Mn}) in nonmagnetic product, and the metallization ratio (MR) of reduced product are chosen as the experimental assessment indicators. The efficiency on reduction and the magnetic separation of manganese ore is evaluated based on the following variables:

$$R_{Mn} = \frac{m_{Mn} \times \beta_{Mn}}{M_{Mn} \times \gamma_{Mn}} \times 100 \quad (1)$$

where R_{Mn} is the recovery ratio of manganese, respectively; m_{Mn} is the mass of the nonmagnetic product; M_{Mn} is the mass of the reduction product; β_{Mn} is Mn grade in nonmagnetic product; and γ_{Mn} is Mn grade in reduction product. Similarly, the iron-removed ratio (R_{Fe}) in magnetic product can be calculated as follows:

$$MR = \frac{MFe}{TFe} \times 100 \quad (2)$$

where MR is metallization ratio of reduced manganese ore; and M_{Fe} and T_{Fe} are the metallic iron content and total iron content in the reduced manganese ore.

Analysis and characterization

The raw ore was divided into two portions: one portion was crushed and milled to sizes smaller than 75 μm for chemical and XRD analysis, whereas the other portion of the reduced samples was used for microstructure of reduced manganese ore. Chemical analyses were conducted at the Metallurgy Research Institute of Northeastern University, Shenyang, China. The XRD analysis of the raw material and the MFS products were obtained using a diffractometer with Cu–Kα radiation (Almelo, the Netherlands). The Cu–Kα radiation (40 kV, 40 mA, wave length 0.154 nm) is used as the X-ray source and the scanned angular range varied from 5° to 90° using a scanning speed of 0.2°/s. The microstructure of the raw material was observed with scanning electron microscopy (SEM, Jena, Germany). In order to determine the relationship between the reaction of carbon and heat and the temperature, the manganese ore and coal mixed are carried out experimentally in thermogravimetric analysis equipped (STA449 F3, Netzsch, Germany). The normal-volume flow of the N_2 carrier gas was found to be 20 cm^3 min^{-1}, controlled by a mass flow meter.

Theoretical fundamental

Major phases in low-grade manganese ore identified by XRD (Fig. 1) are that the manganese and iron mainly comprised of manganite MnO(OH) and bixbyite ($(Fe, Mn)_2O_3$). Figure 2 displays the thermodynamic graph by FACTSAGE 7.0 of directly reduced manganese oxide and iron oxide in low-grade manganese ore. The reduction reaction relationships are calculated as follows:

$$2MnO_2 + C = Mn_2O_3 + CO(g) \quad (3)$$

$$3Mn_2O_3 + C = 2Mn_3O_4 + CO(g) \quad (4)$$

$$Mn_3O_4 + C = 3MnO + CO(g) \quad (5)$$

$$MnO + C = Mn + CO(g) \quad (6)$$

$$3MnO + C = Mn_3C + 3CO(g) \quad (7)$$

$$3Fe_2O_3 + C = 2Fe_3O_4 + CO(g) \quad (8)$$

$$Fe_3O_4 + C = 3FeO + CO(g) \quad (9)$$

$$FeO + C = Fe + CO(g) \quad (10)$$

The above equation shows that MnO_2 and Mn_2O_3 should be reduced to Mn_3O_4 at lower temperature. Further, it is reduced to MnO at above 284°C; the MnO by carbon at atmospheric pressure is reduced to Mn and Mn_3C at above 1393°C and 1407°C, respec-

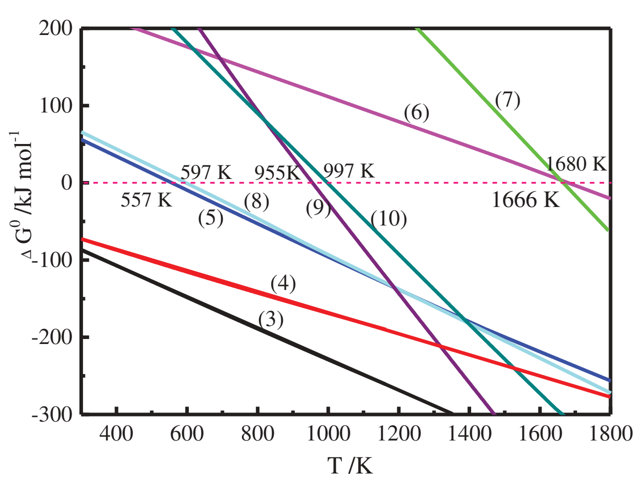


Figure 2. Thermodynamics graph of directly reduced ferromanganese and iron oxide.

tively. However, the MnO is not reduced, and Mn and Mn_3C are not formed at the experiment conditions in this paper. The reduction of Fe_2O_3 by carbon could be feasible at 324°C. Fe_3O_4 reduced is calculated to occur at above 682°C. The metal Fe formed is reduced from FeO at 724°C:

$$2MnO_2+CO(g)= Mn_2O_3+CO_2(g) \quad (11)$$

$$3Mn_2O_3+CO(g)= 2Mn_3O_4+CO_2(g) \quad (12)$$

$$Mn_3O_4+CO(g)= 3MnO + CO_2(g)$$

$$MnO + CO(g)= Mn + CO_2(g) \quad (14)$$

$$C + CO_2(g)= 2CO(g) \quad (15)$$

$$FeO + CO(g)= Fe + CO_2(g) \quad (16)$$

$$Fe_3O_4+4CO = 3Fe + 4CO_2 \quad (17)$$

$$Fe_3O_4+CO(g)= 3FeO + CO_2(g) \quad (18)$$

$$3Fe_2O_3+CO(g)= 2Fe_3O_4+CO_2(g) \quad (19)$$

As shown in Fig. 3, the indirect reductions of manganese oxide and iron oxide mainly can be investigated by thermodynamic graph of manganese and iron oxide. The reduction reaction of MnO_2, Mn_2O_3, and Mn_3O_4 can be converted to MnO at the low temperature. Iron oxide is more difficult to reduce to metal Fe with comparing Mn oxide. The carbothermal reductions of MnO cannot be reduced under reduction temperature 1412°C in the thermodynamics graph. [4,15,24]

Reduction roasting process of samples

The TG and DTG tests are carried out using a thermogravimetric analysis equipped and completely controlled by a PC computer. The samples with a weight of around 20 mg are heated in the temperature interval 50–1000°C with heating rate of 10°C/min. The TG and DTG of manganese ore are shown in Fig. 4. Based on the FC/O ratio of 1.2, the manganese ore and coal mixed are carried out. The weight loss ratio of mixed samples corresponds to first peak in the weight loss rate curve at around 540°C. The reduction reactions of Mn_xO_y are converted to MnO and Fe_3O_4 formed by thermodynamic graph at this temperature. In the temperature range between 751°C and 1000°C, the weight loss ratio arrives at the peak value at around 897°C, which is attributed to that higher valence ferromanganese oxides is reduced to MnO and Fe by carbon gasification and indirect reduction.

Results and discussion

Effects of roasting process on Fe–Mn separation efficiency

The effects of reduction temperature, roasting time duration, the FC/O, ore size on the reduction of manganese ore on the reduction, and Fe–Mn separation efficiencies were determined under the following magnetic separation. This study aimed to investigate the efficiency of Fe and Mn separation from manganese ore via reduction roasting.

Effect of roasting time

The effect of roasting time on efficiency of reduced manganese ore and magnetic separation is shown in Fig. 5. The MR of reduced product increases from 85.74% to 92.60% with an increase of roasting time. As the roasting time increased from 2 h to 6 h, the RFe is slightly improved from 71.66% to 73.59%, while Mn/Fe and

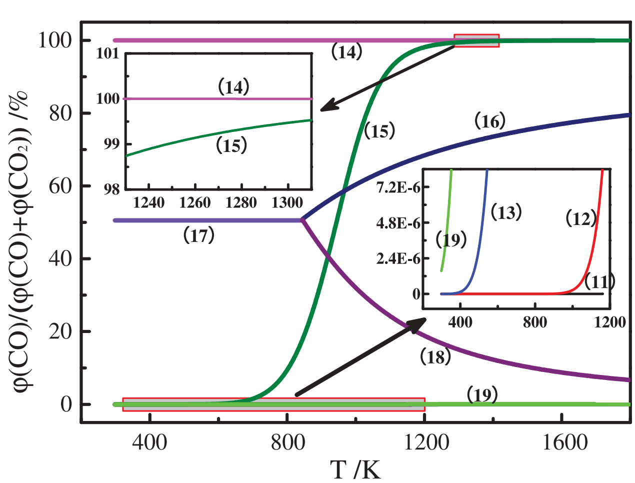


Figure 3. Equilibrium diagram of indirect reduced ferromanganese and iron oxide.

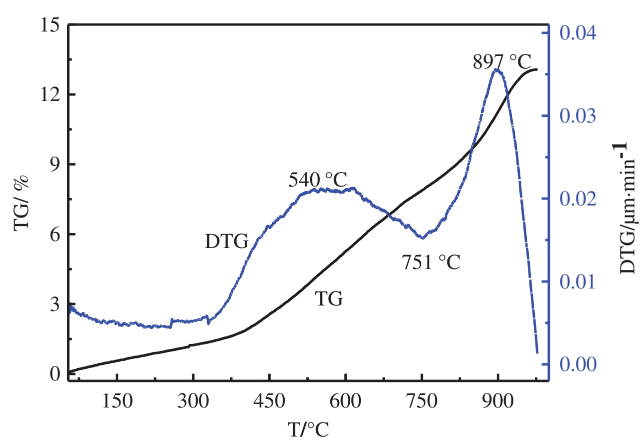


Figure 4. The comparison of TG and DTG curves of mixed manganese ore and coal.

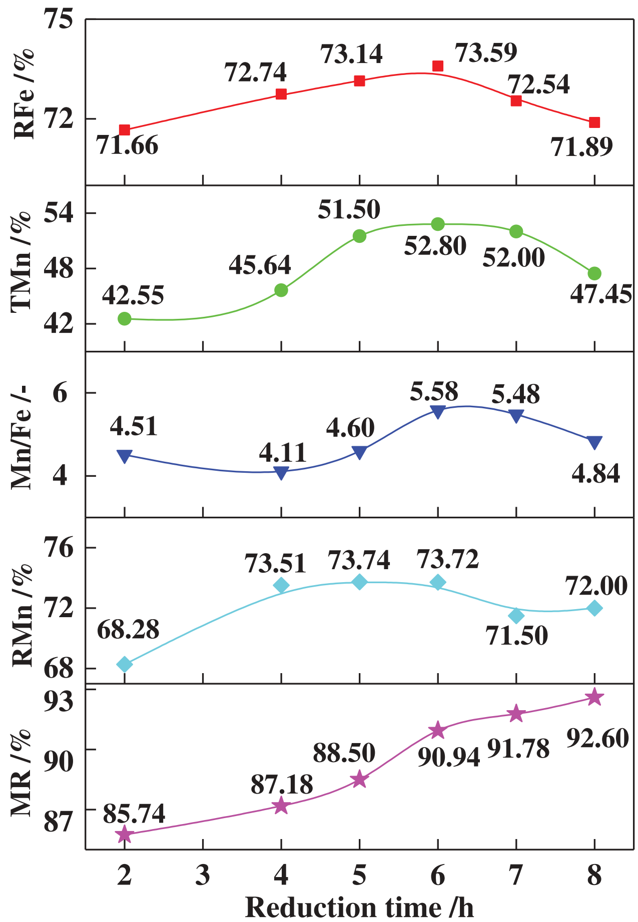


Figure 5. Effect of roasting time on the reduction and the magnetic separation of manganese ore.

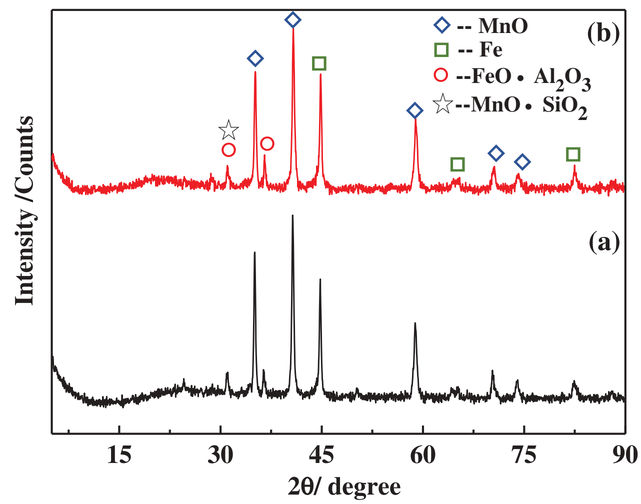


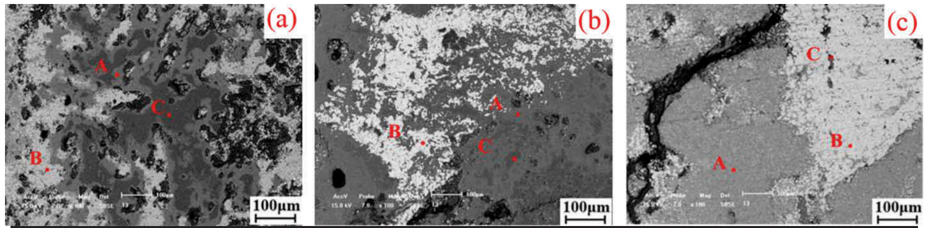
Figure 6. XRD patterns of reduced manganese ore with different roasting time: (a) 2 h and (b) 6 h.

TMn are significantly increased from 4.51 to 5.58 and from 42.55wt% to 52.80wt%, respectively. Meanwhile, the RMn is remarkably increased from 68.28% to 73.72% with roasting time increasing from 2 h to 6 h. When roasting time exceeds 6 h, there are no obvious improvements of TMn of magnetic product. While the RFe, TMn, and Mn/Fe are remarkably decreased to 71.89%, 47.45wt %, and 4.82, respectively. Smelting standard HCFeMn alloy containing minimum 75%Mn requires higher grade manganese in manganese ore with the Mn/Fe ratio of higher than 6.2.[20] Therefore, the assessment indicators of the process are obtained relatively better levels with the roasting time of 6 h, and thereby the roasting time is set as 6 h in the following experiments.

The XRD analysis of reduced manganese ore is shown in Fig. 6. Manganese is mainly present in the form of MnO and manganese silicate ($MnO \cdot SiO_2$) with a little amount in reduced product. Iron is mostly present in the form of metallic iron together with small amount of hercynite. With an increase in the roasting time, there is a little change in the compositions of the phase while the intensity of peaks shows different variations. Although FeO derived from the decomposition of the hercynite is instantaneously reduced to Fe with increasing the roasting time, it is difficult to detect separate FeO phase by the diffraction peak and the intensity of Fe peaks is strengthened. The microstructure of the reduced manganese ore measured by SEM-EDS analysis is shown in Fig. 7. As evident in the figure, the white region is metallic iron, the gray area is unreduced phase, and the black district indicates manganese oxide. Figure 7(a) shows the nucleation and growth of a metallic phase at a roasting time of 2 h. It is revealed that the reduced product presents complex mineralogical structure. The metallic iron and manganese phases are mutually wrapped and inlaid. The iron oxide within the area of gangue was failed to fully decrease. Therefore, it is difficult to separate the iron and manganese by magnetic separation due to the lower MR of reduced manganese ore. With a roasting time of 6 h, the ferromanganese particles are mostly reduced, as seen in Fig. 7(b). With further increasing roasting time from 6 h to 8 h, the accumulation of Fe phase can be seen in Fig. 7(c), and the MnO phase is surrounded by the Fe phase. As a result, the interaction of MnO phase and metallic iron can lead to a difficulty in the separation between iron and manganese and the decrease of RMn and RFe.

Effect of roasting temperature

The effect of reduction temperature on the reduction and the magnetic separation of ferromanganeseore is shown in Fig. 8. It is observed that the MR of reduced manganese ore is dramatically enhanced with increasing reduction temperature. When the reduction



	Point	Mn	Fe	Si	Al	O
(a)	A	43.23	16.61	4.91	8.79	25.30
	B	13.26	81.10	0.34	0.95	4.23
	C	32.45	32.5	3.25	4.26	21.59
(b)	A	42.13	17.71	5.11	8.59	25.30
	B	2.73	92.72	0.34	0.95	3.26
	C	33.36	15.50	12.30	10.05	28.79
(c)	A	45.21	16.21	5.11	5.12	25.3
	B	4.52	89.12	1.21	0.89	2.78
	C	5.50	33.36	52.35	1.21	6.72

Figure 7. SEM-EDS analysis for reduced manganese ore with different roasting time: (a) 2 h, (b) 6 h, and (c) 8 h.

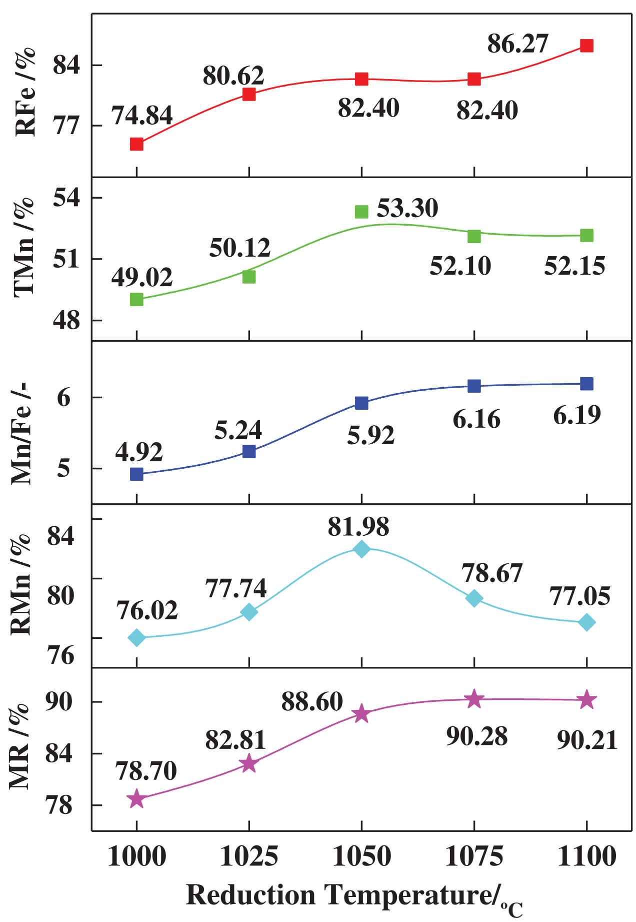


Figure 8. Effect of reduction temperature on reduction and magnetic separation of manganese ore.

temperature is more than 1050°C, it has a small effect on MR. The Mn/Fe in the nonmagnetic product is significantly increased from 4.92 to 5.92 with increasing the reduction temperature from 1000°C to 1050°C, and subsequently level off 6.19 at 1100°C. The efficiency of RFe is correspondingly increased from 74.84% at 1000°C to 86.27% at 1100°C. The TMn and RMn in nonmagnetic product firstly increase and then decrease. With the reduction temperature of 1050°C, RMn and TMn achieve the maximum values of 81.98% and 53.30wt%, respectively. Therefore, the recommended reduction temperature is 1050°C in the subsequent experiments.

Figure 9 shows the iron and manganese oxide distributions in reduced manganese ore at different reduction temperatures. With the reduction temperature of 1000°C, the iron oxide and manganese oxide are gradually reduced and concentrated in the local part of the samples because the chemical composition of manganese ore is non-uniform and thereby the reduction reaction is asynchronous. With increasing the reduction temperature, the fragmented iron rich in the local part increases and aggregates. The excessive enrichment of metallic Fe occurs in certain region with increasing the reduction temperature to 1100°C. As a result, some of the ferromanganese surrounded by metallic iron may also be removed during the magnetic separation process. Therefore, the RMn is lower at 1100°C and the RFe of reduced product is higher.

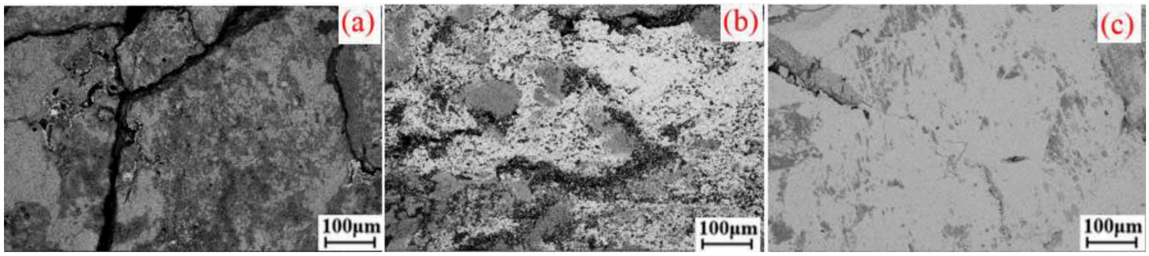


Figure 9. SEM analysis for reduced product at different reduction temperature: (a)1000°C, (b) 1050°C, and (c) 1100°C.

Effect of FC/O

The efficiency of reduced manganese ore and magnetic separation with adding different amounts of reduction coals is shown in Fig. 10. The MR and TMn of reduced manganese ore increase proximately linearly with increasing the FC/O ratio. The Mn/Fe increases from 4.82 to 5.44 with increasing the FC/O ratio. When FC/O ratio is below 2.5, the efficiency of the RFe decreases significantly from 84.09% to 82.75%, and the RMn increases from 70.76% to 76.48%. However, when the FC/O ratio is above 2.5, the RFe and RMn change slowly. Therefore, the FC/O ratio of manganese ore used in the subsequent test is found to be 2.5. With increasing the FC/O ratio, the reduction of iron oxide to metallic iron and high manganese oxide to MnO increased, which leads to the high MR of reduced ore and high RMn in nonmagnetic product.

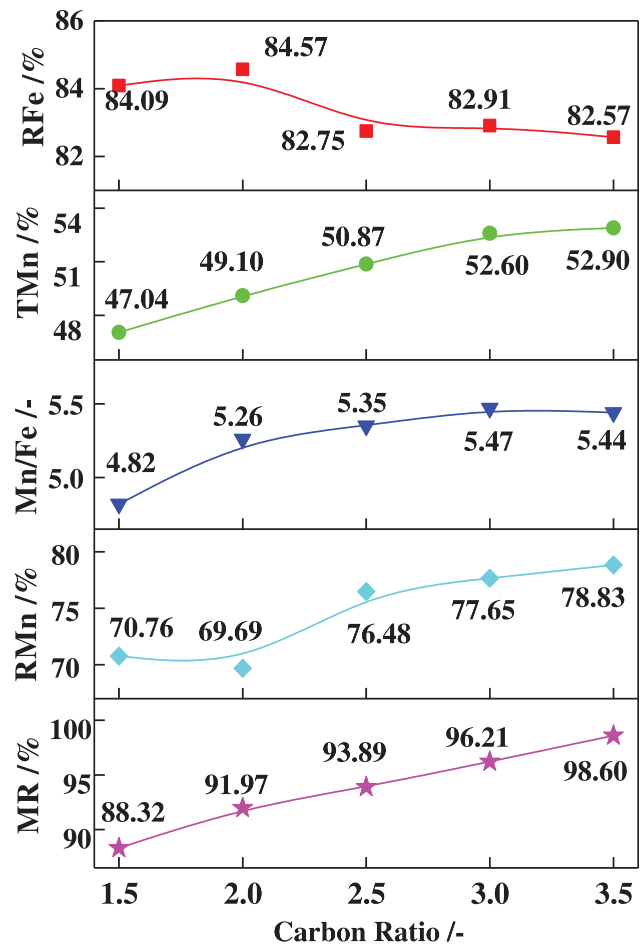


Figure 10. Effect of FC/O on the reduced manganese ore and magnetic separation.

Effect of manganese ore particles size

The effects of the manganese ore particles size on reduced manganese ore and magnetic separation are given in Fig. 11. The results indicate that the RFe decreases with increasing the ore particles size. When the ore particles size is less than 8 mm, a smaller particle size results in a higher MR, and the Mn/Fe in nonmagnetic product changes little, and the RMn and TMn increased significantly from 70.54% to79.96% and from 47.77wt% to 52.10wt%, respectively. When the particles size is above 8 mm, the effects of ore particles size on MR are negligible. The Mn/Fe by magnetic separation dramatically decreases from 5.37 to 3.69 in the experimental range, and the TMn decreases slowly from 52.10wt% to 49.50wt %. When the particle size of ferromanganese ore is 5–8 mm, the TMn and Mn/Fe in nonmagnetic product reach a peak value of 52.10wt% and 5.37, respectively. From the perspective of Mn recovery and smashing costs, Mn recovery reaches to 83.30% and large size minerals are easily obtained by smashing to decline cost. Therefore, the particle size of manganese ore used in the subsequent test is found to be 8–13 mm.

The SEM analysis of reduced manganese ore shows the contrast of outer region and inner region of samples is shown in Fig. 12. It can be seen that reduction process of manganese ore proceeds from the external to the internal. Iron oxide is reduced to metallic iron and manganese oxides to MnO. The RFe decreases and the RMn increases with an increase of ore particles size, which is due to the incomplete reduction of manganese ore. Furthermore, the reduction degree of manganese

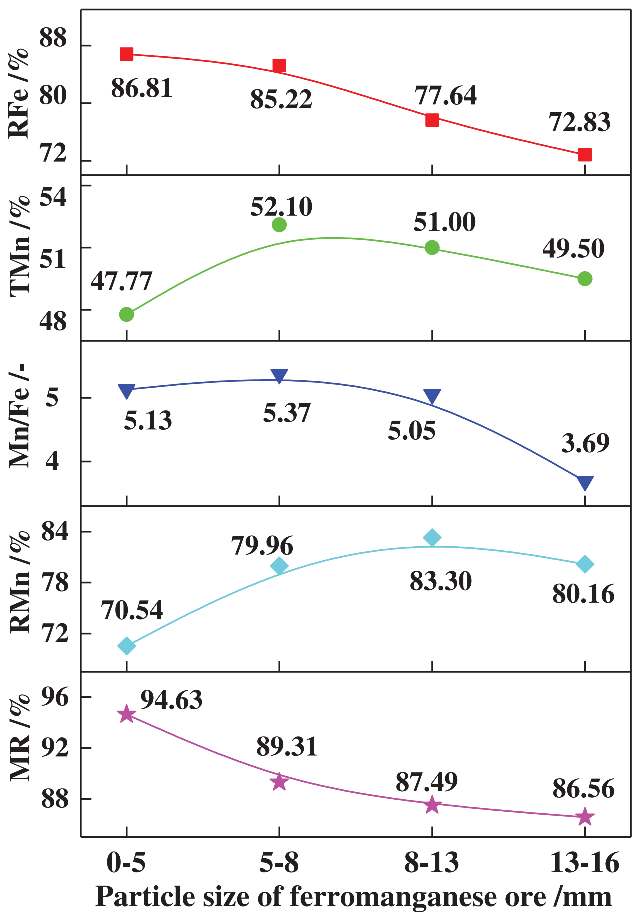


Figure 11. Effect of particles size on the reduced manganese ore and magnetic separation.

ore can be significantly affected by the size of manganese ore. Thus, the bigger particle size results in the lower Mn/Fe and RFe.

Phase transformation of minerals during selective reduction

The main reduction process of manganese ore at certain roasting time can be identified from XRD patterns. Figure 14 integrated by Fig. 13 shows the phase transfer ruler of Mn and Fe compounds in low-grade manganese ore by an easily understandable way, which indicates that the reduction process of manganese ore begins with the reduction of high manganese oxides to MnO and iron oxides to metallic iron.

At first the main manganese oxide phases in manganese ore consists of manganite (MnO(OH)) and iron manganese oxide ((Fe,Mn)$_2$O$_3$). The iron oxide is mostly found in the form of iron manganese oxide and hematite (Fe_2O_3). After 5 min of roasting time, spectra MnO are detectable by XRD analysis. Most of the SiO_2 phase is mainly existed in form of tephroite (MnO· 3Mn_2O_3· SiO_2). Next, Fe_2O_3 phase is disappeared and two new phase (FeO and Fe) generated, but Fe_3O_4 is not detected in XRD spectra, which could be formed and then disappeared or the less its content. When the roasting time is above 60 min, manganese is mainly present as manganese silicate (MnO· SiO_2) and MnO, and iron compounds in form of ferrous oxide are further reduced to metallic iron. Finally, iron is mostly found in the form of metallic iron and a small amount in the hercynite (FeO· Al_2O_3). The reduction of iron oxide is similar to that of previous study.[21,22] The above results indicate that the MnO and metallic iron is mainly existed in the final roasting product. Full reduction of iron and manganese oxides has been achieved during the reduction process. It is mainly consideration that the effect of the process of magnetic separation on improving the recovery rate of Fe and Mn from low-grade manganese ore. The ferromanganese compounds change in the order of Mn_2O_3, MnO· 2Mn_2O_3· SiO_2, MnO· SiO_2 and MnO, and iron compounds change in the order of Fe_2O_3, Fe_3O_4, FeO· Al_2O_3, FeO and Fe with increasing the roasting time.

Magnetic separation of reduction samples

Effects of magnetic separation process on Fe–Mn separation efficiency

Figure 15 shows the effect of magnetic field strength on the efficiency magnetic separation of reduced manganese ore. The Mn/Fe, RFe, and TMn of reduced manganese ore linearly increase with increasing the magnetic field

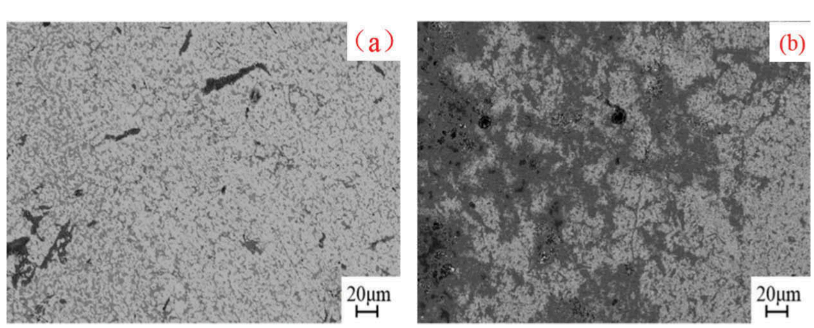


Figure 12. SEM analysis for reduced manganese ore: (a) outer region and (b) inner region.

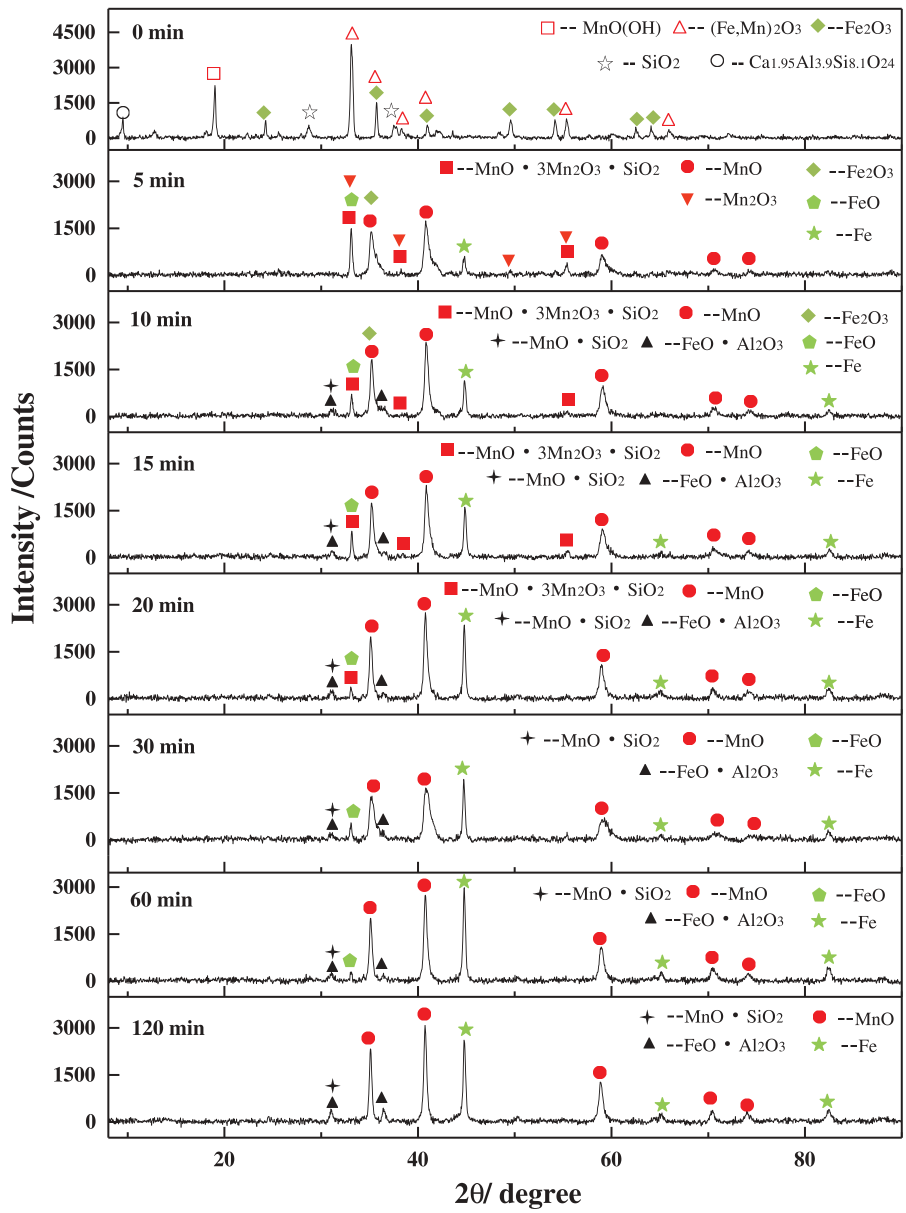


Figure 13. XRD patterns of raw manganese ore at roasting time.

strength. However, the RMn decreases from 85.20% to 71.00%. When the magnetic field strength reaches 100 mT, the RFe and TMn are found to be 93.60% and 56.20wt%, respectively. The Mn/Fe in nonmagnetic product is 6.72 at 50 mT, which can satisfy the production of high-quality ferromanganese alloy.[6]

Analysis of magnetic separation products

Figure 16 shows the SEM-EDS analysis of reduced sample shows that metallic iron and MnO phase mutually interlocked in reduced manganese ore, thus resulting in the decrease of TMn and manganese recovery. It can be seen that each particle has some magnetism. The presence of point A indicates that magnetic iron adheres to the surface of nonmagnetic manganese oxide phase, which deleteriously affects the magnetic separation of iron phase and manganese phase. As shown in point B of Fig. 18, the weak magnetic products may be separated from the reduced product under strongly magnetic field strength, but it could be separated into the nonmagnetic product under weak magnetic field strength. Comparing Fig. 18 with Fig. 17, the metallic iron is entrained and wrapped into the nonmagnetic substance at 25 mT, which leads to a decrease in TMn and Mn/Fe. When magnetic field strength is 100 mT, the iron particles are not found in the non-magnetic material, which leads to a decrease in manganese recovery.

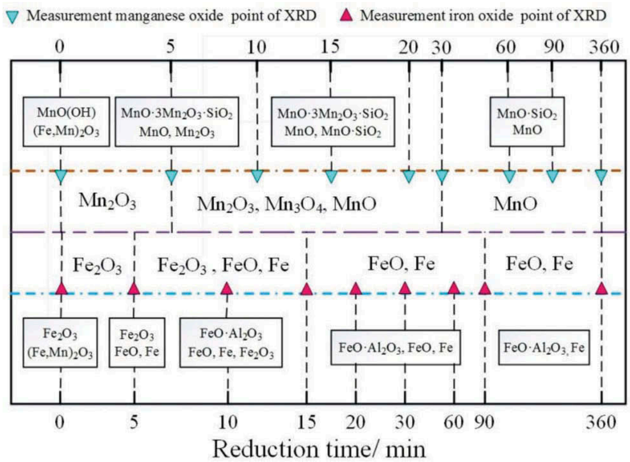


Figure 14. Phase transfer diagram of reduced manganese ore.

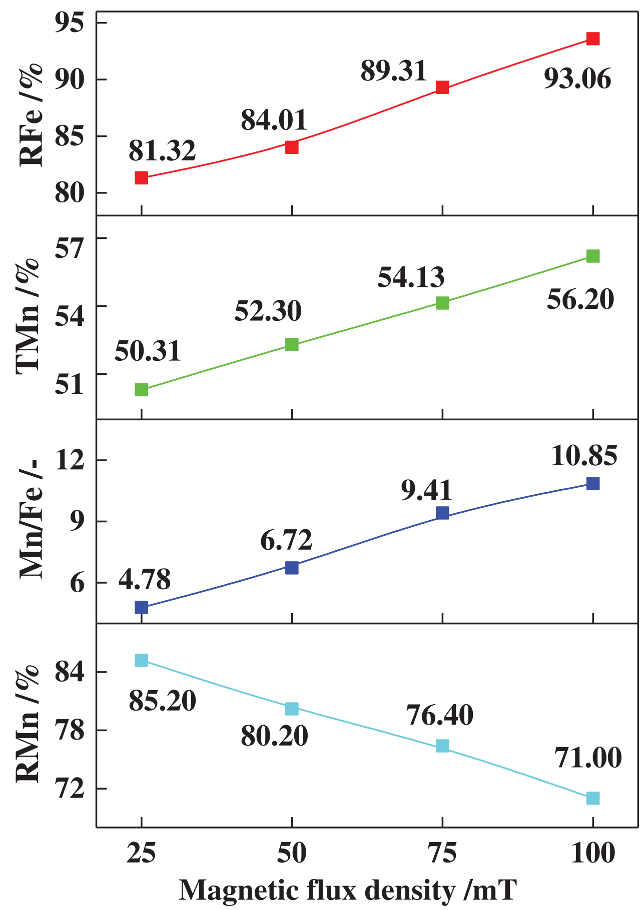


Figure 15. Effect of magnetic flux density on the magnetic separation.

Conclusions

In this study, a cost-efficiently way to upgrade the low-grade manganese ore is successfully beneficiated by reduction roasting followed by magnetic separation. It is also applicable to kinds of low-grade ferromanganese and ferromanganese tailing dumps which beneficiates manganese by reduction roasting and magnetic separation. The manganese oxide and iron oxide in manganese ore should be further reduced by coal, which converted to MnO and metallic iron, respectively.

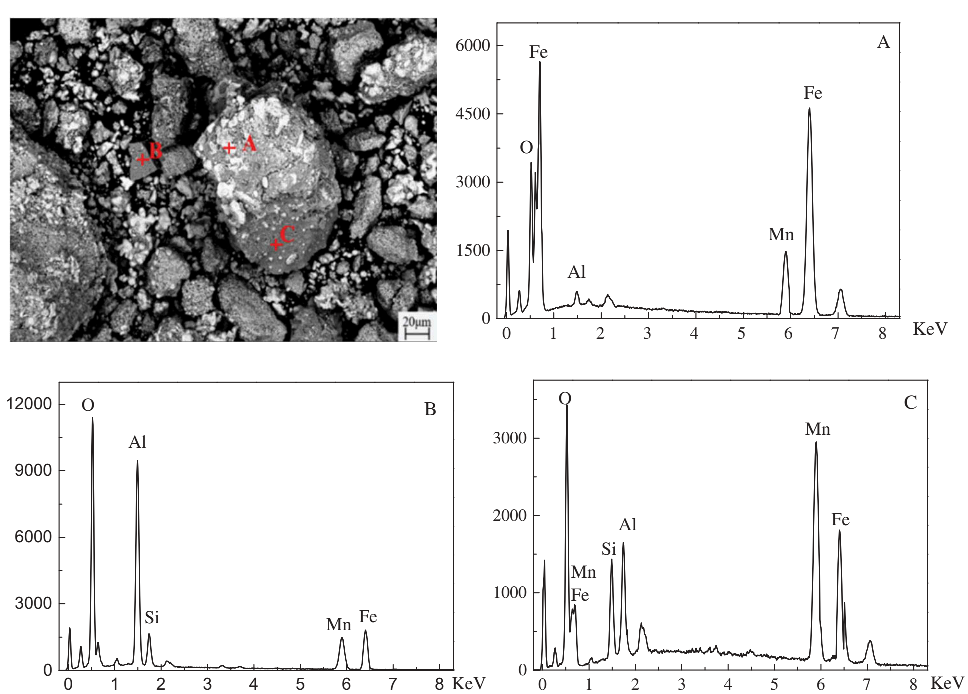


Figure 16. SEM/EDS of reduced manganese ore at 1050°C after 6 h.

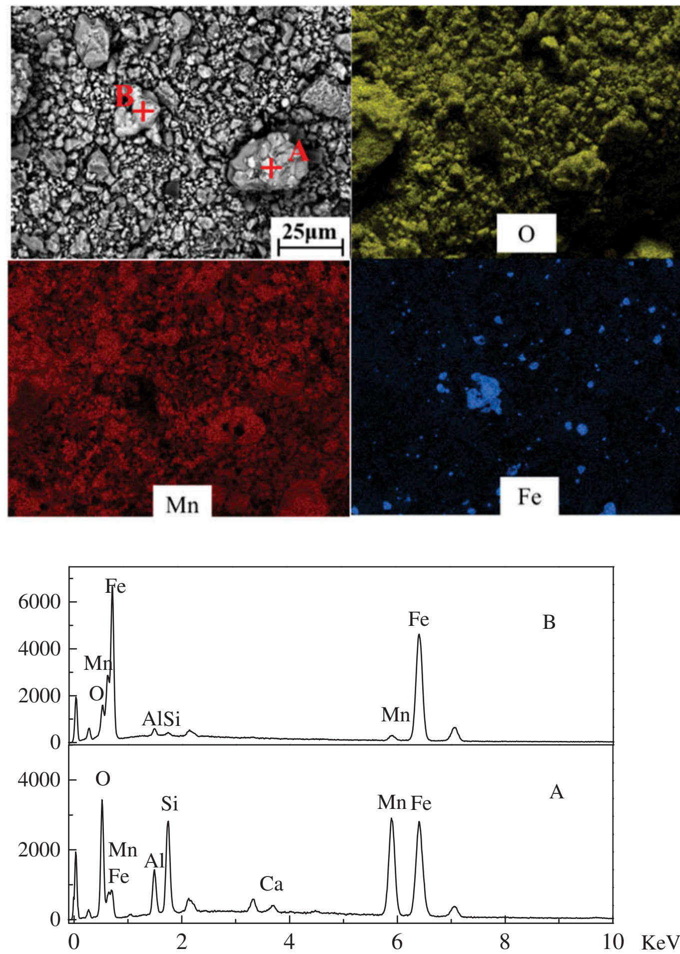


Figure 17. SEM/EDS of nonmagnetic products at magnetic flux density of 25 mT.

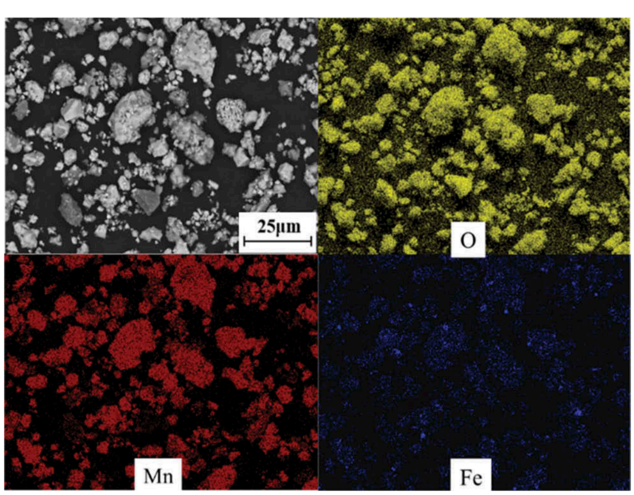


Figure 18. SEM/EDS of nonmagnetic products at magnetic flux density of 100 mT.

(1) The optimized reduction roasting parameters for low-grade manganese ore includes a roasting temperature of 1050°C, roasting time of 6 h, a manganese ore size of 8–13 mm, and an FC/O ratio of 2.5. The reduction roasting products of low-grade manganese ore, mainly comprising of MnO, metallic iron and residual gangue, could be separated effectively to obtain rich-manganese ore and magnetic product (metallic iron). At a rational magnetic field strength of 100 mT, the RMn, RFe, and TMn could reach the peak values of 71.00%, 93.60%, and 56.20wt%, respectively. In particular, based on the abovementioned parameters, the Mn/Fe ratio of the rich-manganese ore is up to 10.85, which meets the requirements of ferromanganese alloy smelting.

(2) Based on the thermodynamics calculations and XRD analysis, the Mn-bearing phase transition could be described as follows: $Mn_2O_3 \rightarrow MnO\cdot 2Mn_2O_3\cdot SiO_2 \rightarrow MnO\cdot SiO_2 \rightarrow MnO$, and the Fe-bearing phases were reduced in the following sequence: $Fe_2O_3 \rightarrow Fe_3O_4 \rightarrow FeO\cdot Al_2O_3 \rightarrow FeO \rightarrow Fe$. Finally, manganese and iron element in the nonmagnetic product

are mostly found in the form of metallic iron and MnO, with a small amount of manganese silicate and hercynite.

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